Solution:

Change into simple integrands

$$\cos 2x = 2\cos^2 x - 1$$

$$\therefore \cos^2 x = \frac{1}{2}[1+\cos 2x]$$

$$\int_0^{\frac{\pi}{2}} \cos^2 x\,dx = \int_0^{\frac{\pi}{2}} \frac{1}{2}[1+\cos 2x]\,dx$$

$$= \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2}\left[\frac{\pi}{2} + 0\right] = \frac{\pi}{4}$$

Note

$e^{a\log x} = x^a$

Example 2.61

Evaluate $\int_0^1 \left[e^{a\log x} + e^{x\log a}\right]dx$

Solution:

$$\int_0^1 \left[e^{a\log x} + e^{x\log a}\right]dx = \int_0^1 \left(x^a + a^x\right)dx$$

$$= \left[\frac{x^{a+1}}{a+1} + \frac{a^x}{\log a}\right]_0^1$$

$$= \left(\frac{1}{a+1} + \frac{a}{\log a}\right) - \left(0 + \frac{1}{\log a}\right)$$

$$= \frac{1}{a+1} + \frac{a}{\log a} - \frac{1}{\log a}$$

$$= \frac{1}{a+1} + \frac{(a-1)}{\log a}$$

Example 2.62

Evaluate $\int_2^3 \frac{x^4+1}{x^2}dx$

Solution:

$$\int_2^3 \frac{x^4+1}{x^2}dx = \int_2^3 \left(x^2 + x^{-2}\right)dx$$

$$= \left[\frac{x^3}{3} - \frac{1}{x}\right]_2^3$$

$$= \left(9 - \frac{1}{3}\right) - \left(\frac{8}{3} - \frac{1}{2}\right) = \frac{13}{2}$$

Example 2.63

Evaluate $\int_{-1}^1 \left(x^3+3x^2\right)^3\left(x^2+2x\right)dx$

Solution:

$$\int_{-1}^1 \left(x^3+3x^2\right)^3\left(x^2+2x\right)dx = \left[\frac{1}{3}\frac{\left(x^3+3x^2\right)^4}{4}\right]_{-1}^1$$

$$\left[\because \int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1} + c\right]$$

$$= \frac{1}{3}(64-4)$$

$$= 20$$

Example 2.64

Evaluate $\int_a^b \frac{\sqrt{\log x}}{x}dx \quad a,b>0$

Solution:

$$\int_a^b \frac{\sqrt{\log x}}{x}dx = \int_a^b (\log x)^{\frac{1}{2}}\frac{dx}{x}$$

$$= \left[2\frac{(\log x)^{\frac{3}{2}}}{3}\right]_a^b$$

$$\left[\because \int [f(x)]^n f'(x)\,dx = \frac{[f(x)]^{n+1}}{n+1} + c\right]$$

$$= \frac{2}{3}\left[(\log b)^{\frac{3}{2}} - (\log a)^{\frac{3}{2}}\right]$$

DO YOU KNOW?

$\log b^{\frac{3}{2}} = \frac{3}{2}\log b$ but,

$(\log b)^{\frac{3}{2}} \neq \frac{3}{2}\log b$ Further,

$(\log b)^{\frac{3}{2}} - (\log a)^{\frac{3}{2}} \neq \left(\log\frac{b}{a}\right)^{\frac{3}{2}}$

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